

# ROI Investment Platform - Technical Documentation

**Version:** 1.0

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**Project:** ROI-based Investment Platform

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## 1. Project Overview

The ROI Investment Platform is a comprehensive investment management system that includes:

- ROI-based investment plans
- Referral income system
- Salary income with qualification rules
- Multi-level referral structure
- Breakdown wallet management
- Integrated deposit/withdrawal systems (USDT and INR)
- Admin, staff, distributor, and white-label systems (B2B/B2C)

### Project Status

| Component | Status   |
|-----------|----------|
| Backend   | Complete |
| Frontend  | Complete |

| Component    | Status                    |
|--------------|---------------------------|
| Docker Setup | Complete                  |
| Database     | Schema & Migrations Ready |
| Seed Data    | Test Accounts Created     |

## 2. Technology Stack

### Architecture Overview

| ROI INVESTMENT PLATFORM                                                    |                                                                                |                                                                                    |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| FRONTEND                                                                   | BACKEND                                                                        | INFRASTRUCTURE                                                                     |
| React 18<br>TypeScript 5.2<br>Vite 5.0<br>Tailwind CSS 3.3<br>Chart.js 4.4 | Node.js 20<br>TypeScript 5.4<br>Express.js 4.19<br>Prisma ORM 5.19<br>JWT Auth | Docker<br>Docker Compose<br>MySQL 8.0<br>Nginx<br>AWS EC2 (Ubuntu 24)<br>Terraform |

## 3. Frontend Technologies

### Core Framework

| Technology | Version | Purpose                 |
|------------|---------|-------------------------|
| React      | 18.2.0  | UI Component Library    |
| TypeScript | 5.2.2   | Type-safe JavaScript    |
| Vite       | 5.0.8   | Build tool & Dev server |

### UI & Styling

| Technology   | Version | Purpose                     |
|--------------|---------|-----------------------------|
| Tailwind CSS | 3.3.6   | Utility-first CSS framework |
| PostCSS      | 8.4.32  | CSS processing              |
| Autoprefixer | 10.4.16 | CSS vendor prefixes         |

### State & Routing

| Technology       | Version  | Purpose                 |
|------------------|----------|-------------------------|
| React Router DOM | 6.20.0   | Client-side routing     |
| Context API      | Built-in | Global state management |

| Technology        | Version | Purpose              |
|-------------------|---------|----------------------|
| AuthContext       | Custom  | Authentication state |
| PermissionContext | Custom  | RBAC permissions     |

## Forms & Validation

| Technology          | Version | Purpose           |
|---------------------|---------|-------------------|
| React Hook Form     | 7.48.2  | Form handling     |
| Zod                 | 3.22.4  | Schema validation |
| @hookform/resolvers | 3.3.2   | Zod integration   |

## Data Visualization

| Technology      | Version | Purpose                    |
|-----------------|---------|----------------------------|
| Chart.js        | 4.4.0   | Charts & graphs            |
| react-chartjs-2 | 5.2.0   | React wrapper for Chart.js |

## HTTP & Utilities

| Technology      | Version | Purpose                |
|-----------------|---------|------------------------|
| Axios           | 1.6.2   | HTTP client            |
| date-fns        | 2.30.0  | Date manipulation      |
| clsx            | 2.0.0   | Conditional classNames |
| react-hot-toast | 2.4.1   | Toast notifications    |

## Frontend File Structure

```

frontend/src/
├── components/
│   ├── AdminLayout.tsx      # Admin dashboard layout
│   ├── Layout.tsx           # User dashboard layout
│   ├── ProtectedRoute.tsx   # Auth-protected routes
│   └── PublicLayout.tsx     # Public pages layout
├── contexts/
│   ├── AuthContext.tsx      # Authentication context
│   └── PermissionContext.tsx # RBAC permissions
├── pages/
│   ├── admin/               # 12 admin pages
│   ├── auth/                # 2 auth pages
│   ├── public/              # 4 public pages
│   └── user/                 # 9 user pages
└── lib/
    └── authStorage.ts        # Token storage utilities

```

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## 4. Backend Technologies

### Core Framework

| Technology | Version     | Purpose              |
|------------|-------------|----------------------|
| Node.js    | 20 (Alpine) | JavaScript runtime   |
| TypeScript | 5.4.5       | Type-safe JavaScript |
| Express.js | 4.19.2      | Web framework        |
| ES Modules | Native      | Module system        |

### Database & ORM

| Technology     | Version | Purpose             |
|----------------|---------|---------------------|
| MySQL          | 8.0     | Relational database |
| Prisma         | 5.19.1  | ORM & query builder |
| @prisma/client | 5.19.1  | Database client     |

### Authentication & Security

| Technology   | Version | Purpose               |
|--------------|---------|-----------------------|
| jsonwebtoken | 9.0.2   | JWT token generation  |
| bcryptjs     | 2.4.3   | Password hashing      |
| Helmet       | 7.1.0   | Security headers      |
| CORS         | 2.8.5   | Cross-origin requests |

### Cryptocurrency & Blockchain

| Technology | Version | Purpose                   |
|------------|---------|---------------------------|
| ethers     | 6.11.1  | Ethereum/EVM interactions |
| tronweb    | 6.0.0   | Tron blockchain           |
| bip39      | 3.1.0   | Mnemonic phrases          |

### Payment Integration

| Technology | Version | Purpose             |
|------------|---------|---------------------|
| Razorpay   | 2.9.3   | INR payment gateway |

### Utilities

| Technology | Version | Purpose            |
|------------|---------|--------------------|
| Zod        | 3.23.8  | Request validation |
| uuid       | 9.0.1   | UUID generation    |
| multer     | 1.4.5   | File uploads       |

| Technology | Version | Purpose               |
|------------|---------|-----------------------|
| morgan     | 1.10.0  | HTTP logging          |
| dotenv     | 16.4.5  | Environment variables |
| node-cron  | 3.0.3   | Scheduled tasks       |

## Backend File Structure

```

backend/src/
├── routes/
│   ├── admin/           # 23 admin API routes
│   ├── webhooks/        # 2 webhook handlers
│   ├── auth.ts          # Authentication routes
│   ├── investments.ts    # Investment routes
│   ├── wallets.ts        # Wallet routes
│   ├── transactions.ts   # Transaction routes
│   └── ... (22 more files)
├── middleware/
│   ├── auth.ts          # JWT authentication
│   ├── adminAuth.ts     # Admin authentication
│   ├── rbac.ts          # Role-based access
│   ├── permissionMiddleware.ts # Permission checks
│   └── security.ts       # Rate limiting, headers
├── services/
│   ├── roiCredit.ts     # ROI credit logic
│   ├── salary.ts         # Salary income logic
│   ├── referral.ts       # Referral commission
│   ├── breakdown.ts      # Breakdown wallet
│   └── ... (15 total services)
├── scripts/
│   ├── seed.ts          # Database seeding
│   ├── cron-roi.ts       # ROI cron job
│   └── cron-salary.ts    # Salary cron job
└── prisma/
    └── schema.prisma     # Database schema (31 models)

```

## 5. Database Schema

### Database: MySQL 8.0 with Prisma ORM

#### Core Models (31 total)

| Model       | Purpose                                          |
|-------------|--------------------------------------------------|
| User        | User accounts with KYC, 2FA, referral codes      |
| Wallet      | Multi-wallet (INR, USDT, ROI, SALARY, BREAKDOWN) |
| LedgerEntry | Wallet transaction ledger                        |
| Transaction | All platform transactions                        |

| Model                | Purpose                            |
|----------------------|------------------------------------|
| RoiPlan              | Investment plans configuration     |
| Investment           | User investments with ROI tracking |
| RoiCredit            | ROI credit history                 |
| Deposit              | INR deposits                       |
| CryptoDeposit        | USDT deposits                      |
| CryptoDepositAddress | Per-user crypto addresses          |
| UserDepositAddress   | ERC20/TRC20 addresses              |
| Withdrawal           | Withdrawal requests                |
| KycSubmission        | KYC applications                   |
| KycDocument          | KYC documents                      |
| KycLog               | KYC audit trail                    |
| Referral             | Referral relationships             |
| ReferralLevelConfig  | Multi-level commission config      |
| ReferralIncome       | Referral earnings                  |
| SalaryRule           | Salary qualification rules         |
| SalaryQualification  | User qualifications                |
| SalaryIncome         | Salary payments                    |
| RoiBoostRule         | ROI boost rules                    |
| RoiBoost             | User boost status                  |
| BreakdownPolicy      | Breakdown deduction policy         |
| BreakdownEntry       | Breakdown wallet entries           |
| RefundRequest        | Refund requests                    |
| Notification         | User notifications                 |
| NotificationTemplate | Notification templates             |
| SupportTicket        | Support tickets                    |
| TicketReply          | Ticket replies                     |
| BlogPost             | Blog posts                         |
| BlogCategory         | Blog categories                    |
| BlogComment          | Blog comments                      |
| CmsPage              | CMS pages                          |
| KbCategory           | Knowledge base categories          |
| KbArticle            | Knowledge base articles            |
| Role                 | Admin roles                        |
| Staff                | Staff accounts                     |

| Model               | Purpose                  |
|---------------------|--------------------------|
| Permission          | RBAC permissions         |
| WhiteLabel          | White label partners     |
| Distributor         | Distributors             |
| DistributorUser     | Distributor-user mapping |
| BrandingSettings    | Platform branding        |
| FeesLimitsSettings  | Fees & limits            |
| ComplianceSettings  | Compliance config        |
| IntegrationSettings | Third-party integrations |
| ChainSettings       | Blockchain settings      |
| CurrencyRate        | Exchange rates           |
| SettingsAudit       | Settings change log      |

## 6. Infrastructure & DevOps

### Docker Configuration

| Component | Image          | Port    |
|-----------|----------------|---------|
| MySQL     | mysql:8.0      | 3306    |
| Backend   | node:20-alpine | 5000    |
| Frontend  | nginx:alpine   | 80/3000 |

### Docker Features

- Multi-stage builds for frontend
- Health checks for MySQL
- Volume persistence for database
- Bridge network for container communication
- Environment variable injection

### AWS Infrastructure (Terraform)

| Resource       | Type         | Details                        |
|----------------|--------------|--------------------------------|
| Provider       | AWS          | Version ~> 5.0                 |
| AMI            | Ubuntu 24.04 | Noble (x86_64)                 |
| Instance       | EC2          | Configurable type              |
| Security Group | VPC          | Ports: 22, 80, 443, 3000, 5000 |
| Storage        | gp3          | 20GB root volume               |

## CI/CD

| Tool           | Purpose                 |
|----------------|-------------------------|
| GitHub Actions | Deployment workflow     |
| Terraform      | Infrastructure as Code  |
| Docker Compose | Container orchestration |

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## 7. Security Features

### Authentication

| Feature            | Implementation                   |
|--------------------|----------------------------------|
| JWT Tokens         | Access + Refresh tokens          |
| Password Hashing   | bcryptjs (salt rounds)           |
| OTP Verification   | SHA256 hashed OTPs               |
| 2FA Support        | Scaffolded (isTwoFAEnabled flag) |
| Session Management | Token expiry & refresh           |

### Security Middleware

| Feature            | Implementation                      |
|--------------------|-------------------------------------|
| Rate Limiting      | In-memory store (Redis recommended) |
| Helmet             | Security headers (XSS, HSTS, etc.)  |
| CORS               | Configurable origins                |
| IP Whitelist       | Optional IP filtering               |
| Request Size Limit | Configurable payload limits         |
| API Key Validation | White label access                  |

### Security Headers

X-Content-Type-Options: nosniff  
X-Frame-Options: DENY  
X-XSS-Protection: 1; mode=block  
Strict-Transport-Security: max-age=31536000; includeSubDomains

### RBAC (Role-Based Access Control)

| Feature           | Implementation                  |
|-------------------|---------------------------------|
| Roles             | super_admin, admin, staff, user |
| Permissions       | Resource + Action based         |
| Staff Departments | FINANCE, KYC, SUPPORT, CONTENT  |

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## 8. Blockchain & Payment Integrations

### Cryptocurrency Support

| Feature    | Technology | Network                   |
|------------|------------|---------------------------|
| ERC20 USDT | ethers.js  | Ethereum                  |
| TRC20 USDT | TronWeb    | Tron                      |
| HD Wallet  | bip39      | Mnemonic-based derivation |

### HD Wallet Implementation

Derivation paths:

- ERC20: m/44'/60'/0'/0/{userIndex}
- TRC20: Same key, converted to Tron address

### Payment Gateway

| Provider | SDK            | Currency |
|----------|----------------|----------|
| Razorpay | razorpay@2.9.3 | INR      |

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## 9. Feature Verification Report

### Implemented Features (Done)

#### *User Features*

- User Registration & Authentication (JWT)
- KYC Verification System
- Dual Wallet System (INR & USDT)
- Investment Plans with ROI Automation
- Multi-level Referral System
- Salary Income System
- ROI Boost (50+ referrals)
- Breakdown Wallet & Refunds
- Withdrawal System
- Transaction History
- Currency Converter
- Notifications
- Support Tickets
- Profile Management

#### *Admin Features*

- Dashboard with Analytics
- User & KYC Management

- Deposit & Withdrawal Approval
- ROI & Salary Management
- Referral System Configuration
- Settings Management
- Staff & Role Management (RBAC)
- White Label System
- Blog & CMS

## Partially Implemented / Scaffolded

| Feature                 | Status     | Notes                       |
|-------------------------|------------|-----------------------------|
| 2FA (Two-Factor Auth)   | Scaffolded | User.isTwoFAEnabled exists  |
| Email/SMS sending       | Stubbed    | Integration settings exist  |
| Video KYC               | Partial    | DocType supports VIDEO      |
| Live currency rates API | Stubbed    | Binance/Moralis integration |
| Export CSV/PDF reports  | Partial    | Some functionality limited  |

## Not Found / May Be Missing

| Feature            | Status    |
|--------------------|-----------|
| E-PIN System       | Not Found |
| QR Code generation | Not Found |
| Tutorial Videos    | Not Found |
| Live Chat (AI bot) | Not Found |

## Implementation Summary

| Category       | Implemented    | Partial     | Missing    |
|----------------|----------------|-------------|------------|
| User Features  | ~90%           | ~8%         | ~2%        |
| Admin Features | ~85%           | ~10%        | ~5%        |
| Integrations   | ~30%           | ~50%        | ~20%       |
| <b>Overall</b> | <b>~80-85%</b> | <b>~12%</b> | <b>~5%</b> |

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## 10. Future Tech Stack & Integrations

### Email & Communication

| Technology  | Status     | Notes                    |
|-------------|------------|--------------------------|
| AWS SES     | Scaffolded | Provider configured      |
| SMTP        | Scaffolded | smtpClient.ts exists     |
| SMS Gateway | Scaffolded | SMS provider in settings |

| Technology         | Status  | Notes                      |
|--------------------|---------|----------------------------|
| Push Notifications | Planned | NotificationTemplate ready |

## Payment & Crypto (Needs Configuration)

| Technology  | Status     | Requirement            |
|-------------|------------|------------------------|
| Razorpay    | Scaffolded | Add API keys           |
| Moralis     | Scaffolded | Add API key + webhooks |
| Binance API | Planned    | Currency rates         |

## Security & Auth

| Technology | Status     | Notes                      |
|------------|------------|----------------------------|
| 2FA (TOTP) | Scaffolded | User.isTwoFAEnabled exists |
| reCAPTCHA  | Scaffolded | Provider configured        |
| Video KYC  | Partial    | DocType supports VIDEO     |

## Cloud Storage & CDN

| Technology   | Status     | Notes                |
|--------------|------------|----------------------|
| AWS S3       | Planned    | fileKey fields ready |
| File Uploads | Scaffolded | multipart installed  |

## Monitoring & Caching (Recommended)

| Technology     | Purpose                          |
|----------------|----------------------------------|
| Redis          | Rate limiting, sessions, caching |
| Sentry/DataDog | Application monitoring           |
| Winston        | Logging                          |

## Configuration Checklist

These integrations need API keys/secrets to work:

1. **Razorpay** → RAZORPAY\_KEY\_ID and RAZORPAY\_KEY\_SECRET
  2. **Moralis** → MORALIS\_API\_KEY and webhook setup
  3. **SMTP** → SMTP\_HOST, SMTP\_USER, SMTP\_PASS
  4. **AWS SES** → AWS credentials
  5. **reCAPTCHA** → RECAPTCHA\_SITE\_KEY and SECRET
-

## 11. API Structure

### Public Routes

- POST /api/auth/register - User registration
- POST /api/auth/login - Login
- POST /api/auth/refresh - Token refresh
- GET /api/health - Health check

### User Routes (Protected)

- /api/wallets/\* - Wallet operations
- /api/investments/\* - Investment management
- /api/transactions/\* - Transaction history
- /api/referrals/\* - Referral system
- /api/profile/\* - Profile management
- /api/kyc/\* - KYC submission
- /api/support/\* - Support tickets
- /api/notifications/\* - User notifications

### Admin Routes (Protected + RBAC)

- /api/admin/dashboard/\* - Dashboard stats
- /api/admin/users/\* - User management
- /api/admin/kyc/\* - KYC approval
- /api/admin/deposits/\* - Deposit management
- /api/admin/withdrawals/\* - Withdrawal approval
- /api/admin/roi/\* - ROI management
- /api/admin/salary/\* - Salary rules
- /api/admin/referrals/\* - Referral config
- /api/admin/settings/\* - Platform settings
- /api/admin/staff/\* - Staff management
- /api/admin/whitelabel/\* - White label
- /api/admin/blog/\* - Blog management
- /api/admin/cms/\* - CMS pages

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## 12. Quick Reference

### Start Development

# Local development  
docker-compose up -d

### # Access URLs

Frontend: `http://localhost:3000`  
Backend: `http://localhost:5000`  
API: `http://localhost:5000/api`

### Test Accounts

| Role  | Email         | Password  |
|-------|---------------|-----------|
| User  | user@roi.com  | User123!  |
| Admin | admin@roi.com | Admin123! |

### Project Metrics

| Metric              | Value     |
|---------------------|-----------|
| Total Lines of Code | 20,000+   |
| Database Models     | 31        |
| Backend Routes      | 46+ files |
| Frontend Pages      | 27        |
| Services            | 15        |
| Middleware          | 5         |

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